SNEHASHIS CHATTERJEE

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EDUCATION

Indian Institute of Technology, Kharagpur (IIT Kharagpur) *Jul 2026 – Jul 2030 (Expected)*
PhD (Working Professional), Electronics and Electrical Communication Engineering (E&ECE). Under Prof. Debashis Sen & Prof. Subhadip Mukherjee

UCST, University of Calcutta, Kolkata *Aug 2016 – Oct 2020*
Bachelor of Engineering, Electronics and Communication Engineering. Overall CGPA: 8.27

CAREER OBJECTIVE

Dynamic and results-driven Senior Research Engineer with deep expertise in computer vision, specializing in advanced algorithms for image processing, object detection, and machine learning. Proven experience in driving research projects from concept to deployment, with strong analytical and theoretical foundations. Adept at collaborating across disciplines to deliver impactful, research-backed solutions. Currently pursuing a Working Professional PhD at IIT Kharagpur, furthering research at the intersection of computer vision and deep learning. Committed to continuous learning and contributing meaningfully to innovative, high-impact projects across academia and industry.

EXPERIENCE

Videonetics Technology Private Limited *India*
Senior Research Engineer *Dec 2025 – Present*

- Developing and deploying advanced AI systems for large-scale video analytics, including computer vision and multimodal models, along with training and fine-tuning Small Language Models (SLMs) and Large Language Models (LLMs) for natural language-driven video search.
- Building high-performance inference and analytics pipelines using FastAPI-based backend services, optimized deployment frameworks such as vLLM and TensorRT-LLM, and hardware-accelerated video processing using OpenVINO, CUDA, GStreamer, and Intel DLStreamer across heterogeneous platforms (CPU, GPU, iGPU).
- Architected a large-scale object re-identification (Re-ID) system for CCTV-based surveillance networks, enabling robust cross-camera matching of persons and vehicles across non-overlapping camera views using appearance-embedding and metric-learning techniques for city-scale, multi-camera tracking.
- Designed, built, and spearheaded video classification models for human behavior analysis tailored to banking customers, detecting behaviors such as frisking pattern analysis to prevent unauthorized entry, counter-visit interactions, and suspicious activity to support in-branch monitoring and customer analytics.
- Working on thermal video analytics to detect moving vehicles and humans, where generic pretrained detectors are hard to train reliably due to the lack of large-scale labeled thermal datasets – adopted and adapted DeepFTSG as the foreground-detection backbone, built a custom UI, and trained the model to produce robust, production-grade results; tackled heavy camera shake from windy outdoor conditions by building a data augmentation pipeline that synthetically injects camera-shake jitter into the limited available footage; and deployed the pipeline on a zero-copy, fully GPU-resident NVIDIA DeepStream architecture combining background-modeling (BGM) with the NvOF optical-flow object tracker alongside custom video-analytics business logic, engineered for maximum FPS and channel count on 10 fps, 720p input streams.

- Led the AI analytics build for a large enterprise safety and security deployment. Designed and shipped 40+ video analytics across workplace safety (slip/fall, PPE, hygiene, blocked fire exits, electrical panel hazards), security (asset carry-out, object removal, two-person rule, weapons) and operations (meal headcount, wrong parking, idle equipment), running on a vision-language model (Qwen3-VL-8B on vLLM) plus classic CV pipelines.
- Built a Python analyzer engine for vision-language models (VLMs): questions go out as clips or single frames with structured yes/no answers, and an event fires only after a configurable run of consecutive "yes" answers in camera time. Prompts hot-reload and are versioned. LoFTR + RANSAC homography handle object-removal and equipment-position checks; SigLIP handles attribute checks (PPE, hairnet, gloves, carry-out).
- Built a natural-language video search assistant from four small models with no LLM call at query time: DistilBERT models for intent routing, attribute extraction and time parsing, plus a summarizer fine-tuned with LoRA from Qwen2.5-0.5B. Held-out results: 99.3% intent accuracy and 0.998 F1 on 11-slot attribute extraction; runs on CPU or about 3.4 GB of GPU.
- Built predictive and multi-camera analytics: a camera Heat-Risk alert forecast backtested on a 7-day holdout, convoy tracking from license-plate (ANPR) events, and VIP movement support. Delivered with the Go backend and web UI as an offline, air-gapped Docker package for a single 32 GB GPU.

Indian Institute of Technology, Madras *Chennai, India*
Project Associate under Prof. Kaushik Mitra *Feb 2025 – Nov 2025*

- Working on SMPL-based models for 3D human mesh recovery, focusing on pose and shape estimation from both monocular and multi-camera inputs.
- Implementing model fitting pipelines, optimizing body model parameters, and integrating vision-based priors for robust reconstruction.
- Contributing to applications such as yoga posture analysis and human motion understanding.

Videonetics Technology Private Limited *Kolkata, West Bengal*
Research Engineer *Apr 2021 – Jan 2025*

- Led the design, development, and large-scale deployment of advanced computer vision-based video analytics solutions for intelligent traffic management, real-time monitoring, and biometric facial recognition systems.
- Developed and trained deep learning classifiers, detection, and segmentation models tailored for diverse video analytics scenarios, ensuring high robustness and accuracy.
- Deployed production-grade solutions across 300+ surveillance cameras, consistently achieving accuracy above 90% under real-world conditions.
- Architected efficient deep learning pipelines optimized for multi-platform deployment on GPU and CPU using FP32, FP16, and INT8 quantization with TensorRT, ONNX, and OpenVINO to enable low-latency inference.

BrainSightAI *Remote (Bangalore, India)*
Research and Development Intern (Machine Learning) *Dec 2020 – Mar 2021*

- Worked on a structural MRI-based glioma analysis pipeline with a focus on large-scale preprocessing of sMRI data.
- Integrated two CNN-based segmentation models (one for skull stripping, one for glioma segmentation) into a unified end-to-end pipeline.

- Enabled efficient and accurate automated analysis through robust model integration and pipeline optimization.

Airprobe *Remote (Bangalore, India)*
Computer Vision Intern *Dec 2020 – Mar 2021*

- Developed a thermal image classifier to identify images requiring normalization.
- Implemented a Detectron2-based model for detecting solar panels of varying shapes under diverse environmental conditions.
- Improved image processing efficiency and detection accuracy through robust model design.

MENTORSHIP

Videonetics Technology Private Limited *Intern Mentor*

- Regularly mentored summer interns, guiding students from RKMVERI (Ramakrishna Mission Vivekananda Educational and Research Institute) and Presidency University through computer vision and deep learning projects.
- Provided hands-on guidance on research methodology, coding practices, and deployment of models into production pipelines.

FrameBengal *Project Mentor*

- Mentored two students on the project "Low-Light Image Relighting under Arbitrary Illumination" – one from Jalpaiguri Government Engineering College and another from University Institute of Technology, Burdwan (University of Burdwan).
- Guided the students through problem formulation, literature review, and experimentation for low-light image enhancement and relighting.

EXTRACURRICULAR ACTIVITIES

Adani India SOAR Hackathon, Computer Vision Track *Adani Corporate HQ, Shantigram*
Representing Videonetics Technology Private Limited

- Represented Videonetics at the Adani India SOAR Hackathon in the Computer Vision domain, competing on-site at Adani's corporate headquarters in Shantigram.
- Tackled a problem statement centered on airport retail analytics, with the team required to design and pitch a complete, workable solution within a 6-hour sprint, backed by a presentation.
- Delivered a real-time video analytics platform built with a scalable, modular architecture – covering stream ingestion, inference, and dashboarding – under tight time constraints.
- Following the hackathon, Adani Airports issued a Letter of Intent to our team for the proposed solution.

PROJECTS

Enterprise Video Intelligence Platform (via Videonetics) *Case Study*
Qwen3-VL, vLLM, SigLIP, LoFTR, DistilBERT, LoRA, DeepStream, Go, Docker

- 40+ production video analytics, a VLM analyzer engine, a natural-language video search assistant and predictive risk forecasting, delivered as an offline single-GPU package.

Multi-view 3D Human Pose Estimation *Google Drive*

- Developed a multi-view 3D human pose estimation pipeline using a four-camera synchronized setup.

- Estimated camera intrinsic and extrinsic parameters through multi-view calibration.
- Reconstructed 3D joint locations via geometric triangulation from multi-view 2D keypoints.
- Utilized AlphaPose for robust 2D pose detection across all camera views.

Thermal Image Super-Resolution and Novel View Synthesis *Google Drive*

- Developed thermal image super-resolution methods using multimodal thermal and RGB data fusion.
- Explored novel view synthesis using 3D Gaussian Splatting (3DGS) and Neural Radiance Fields (NeRF).
- Investigated sparse-view novel view synthesis to improve reconstruction quality under limited viewpoints.

PUBLICATIONS & PATENTS

System and Method for Tracker-Lifecycle-Bounded Localization Scheduling
Indian Patent (Status: Filed). Inventors: Dr. Sudeb Das, Snehashis Chatterjee, Tuhin Bose.

COURSES & TECHNICAL EXPERTISE

Graduate-Level Courses

- **Computational Photography (EE5176)**, IIT Madras Grade: 10/10
- **Deep Learning for Imaging (EE5179)**, IIT Madras Grade: 7/10

Online Certifications (NPTEL)

- **Introduction to Machine Learning**, IIT Kharagpur, Prof. Sudeshna Sarkar Grade: 79%
- **Digital Image Processing (Elite)**, IIT Kharagpur, Prof. Prabir Kumar Biswas Grade: 73%

Core Domains

- Computer Vision, Digital Image Processing, Machine Learning
- Computational Photography, 3D Vision, Human Pose & Shape Estimation

Deep Learning & ML Frameworks

- PyTorch, OpenCV, TensorRT, OpenVINO, ONNX, vLLM, TensorRT-LLM, Detectron2

Programming

- Python, C++, FastAPI, GStreamer, CUDA